

# DEEP LEARNING FOR FINANCE

## Project # 2

### Scenario

Most institutional investors' portfolio allocations have exposure to multiple asset classes and/or geographical areas. This makes a lot of sense, since this way investors can exploit variations in rewards for risk across different asset classes throughout market regimes. Strategic Asset Allocation (SAA) and Tactical Asset Allocation (TAA) are different approaches to multi-asset investment strategies. You can read more about these [in this short article from Vanguard](#).

In this GWP, you will work on producing a Deep Learning model that tries to predict short-term market trends across different asset classes. Then, you will test the out-of-sample performance of different asset allocation strategies based on the predictions of the model.

### Tasks

#### Step 1

First, you need to gather data on the different asset classes that we are going to consider. To simplify, in this case we will work with 5 different ETFs, each a major representative of their asset class. Here you have their symbols, so that you can work with the **yfinance** library to download the data:

- Equity → SPDR S&P500 ETF. Symbol: "SPY"
- Fixed Income → iShares 20+ Year Treasury Bond ETF. Symbol: "TLT"
- Cash-like → iShares 1-3 Year Treasury Bond ETF. Symbol: "SHY"
- Precious metals (Gold) → SPDR Gold Shares. Symbol: "GLD"
- Crude Oil → Invesco DB Oil Fund. Symbol: "DBO"

- a. First, use the **yfinance** library to download the data. You need to **get daily prices** for each of the 5 ETFs. You will use data in the period from January 1<sup>st</sup>, 2018 to December 30<sup>th</sup>, 2022 for your **test sample**. We are leaving to your discretion the definition of validation and training sample periods.
- b. Once you have downloaded the data for each individual ETF, **perform some exploratory data analysis**. Specifically, use visualization tools to describe the different dynamics and characteristics of the time-series (summary statistics, seasonality, stationarity, ...). Try keeping this brief and focus on the main characteristics you consider worth it of highlighting. You can focus on prices and/or daily returns. The exploratory data analysis should address both individual distributions AND joint distributions, including correlations/covariances.

## Step 2

In Step 2, you will create a Deep Learning model using recurrent neural networks to predict the 25-day ahead return of each of the 5 different asset classes.

- a. For each of the 5 different asset classes, create a Deep Learning model that uses LSTM network architecture (you may combine LSTM layers with regular dense layers) to **predict the 25-day ahead return of each ETF**. Use as inputs the past information of the time series (feel free to include whatever lags in returns you consider appropriate). You are free to select the model architecture you like. You can even play around with different architectures and choose the one that performs better.
- b. **Train** each of the 5 models. Comment on the in-sample predictive performance of the models, as well as the comparative results across asset classes.
- c. **Test** each of the 5 models **out of sample**. Comment on the out-of-sample predictive performance of the models, as well as the comparative results across asset classes.
- d. **Create a trading strategy that uses the out-of-sample predictions of the 5 different models**. You can choose the strategy that you want. You could, for example, decide to go long on the 2 (or 3) asset classes the model predicts will do best and go short on the 2 (or 3) asset classes the model expects to do

worse. **The strategy needs to be rebalanced every 25 days and use information from the predictions of at least 4 of the 5 asset classes.**

- e. **Backtest** the previous strategy in the test period. Comment on the results. Compare the results versus the Buy-and-hold return of an Equally weighted portfolio of the 5 asset classes.

### Step 3

So far, our models have always had 1 output unit. You will now implement a Deep Learning model with multi-output. If you need more information on multi-output regression, feel free to research online.

- a. Create a **multi-output model** that blends together all of the information from the 5 models of Step 2. Specifically, create a model architecture that uses all the inputs from the 5 different previous models together to predict the 25-day ahead return from the 5 different ETFs. In other words, the model will have 5 outputs (one for each 25-day ahead return of an ETF).
- b. **Train and test** the previous model. Comment on the in- and out-of-sample predictive performance of the model and compare them with the ones from Step 2.
- c. **Create a trading strategy that uses the out-of-sample predictions of the model.** Same instructions as in Step 2 part d) apply.
- d. **Backtest** the previous strategy in the test period. Comment on the results. Compare the results versus the (i) Buy-and-hold return of an Equally weighted portfolio of the 5 asset classes, and (ii) the return from the trading strategy of Step 2 part e).

### Step 4

Now, as a group, discuss the different results obtained. Specifically, what are the predictability implications of the multi-output versus the single-output architectures? What about in terms of backtesting performance? Is the same information being captured by both model architectures?

### Step 5

As a group, please submit all files that reproduce the assignment. This will include the Python file for all the steps as well as work together on a report that contains the answers to all the questions above (except for the code) in a clear and organized manner (think about how you would present this to your boss).